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Frank Angelo Malubag

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Summary

Full-Stack Backend-focused Software Engineer and **DataCamp-certified AI Engineer Associate** building web, cloud, and LLM-powered applications. Built and deployed full-stack systems, including an HR management system used by a local municipality. Skilled in Go, TypeScript, Python, Node.js, PostgreSQL, Docker, Terraform, AWS, and LLM application development, including prompt engineering, embeddings, semantic search, RAG, and tool-calling agents. **Former Web3 Operations Manager** and **customer service retention specialist**, bringing leadership experience, technical communication, and a security-first engineering mindset.

Skills

Languages: JavaScript, TypeScript, Go, Rust, Python, SQL
Frontend: React, Next.js, HTML, CSS, TailwindCSS
Backend: RESTful APIs, Node.js, Express.js, PostgreSQL, MongoDB, SQLx, Actix Web
AI Engineering: LLM Application Development, Prompt Engineering, Embeddings, Semantic Search, RAG, LLM Evaluation, LLMOps
LLM Frameworks & Tools: OpenAI API, Anthropic API, Hugging Face, LangChain, LangGraph, Chains, Pipelines, Prompt Templates
Retrieval, Agents & Protocols: Pinecone, ChromaDB, Vector Search, Similarity Search, AI Agents, ReAct, Tool Calling, MCP
Architecture & Messaging: Microservices, Event-Driven Architecture, Pub/Sub, RabbitMQ Fanout, MVC
Cloud & Infrastructure: Docker, Docker Compose, Terraform, AWS, Cloudflare R2, K3s, RabbitMQ, GitHub Actions
ML & Data: NumPy, Pandas, PyTorch
Engineering Practices: Software Design Principles, Testing, CI/CD, Security Hardening, Observability, Git, Agile
Soft Skills: Team Leadership, Mentoring, Technical Communication, Negotiation, Conflict Resolution, Self-Directed Learning, Ownership

Work Experience

Independent Software Engineer
Jan 2024 - Present
Self-Employed
Philippines (Remote)

- **Self-directed study** in backend engineering, distributed systems, and AI & ML, applied by building production-style systems solo.
- Earned the **DataCamp AI Engineer for Developers Associate** certification (Nov 2026), covering LLM application development, prompt engineering, RAG, and vector databases.
- Member of the **NVIDIA 6G Developer Program** since Mar 2026; built **Sionna Neural Receiver**, a PyTorch 6G receiver prototype.
- Built a LinkedIn network of tech industry leaders, including **CTOs and CEOs**, while publishing regular tech content.

Software Developer Apprentice
Nov 2023 - Dec 2023
Atlas Analytics
Singapore (Remote)

- Engineered a **report generation automation workflow** that aggregated user-specific marketing analytics through Xano APIs, rendered them in WeWeb, and produced PDF-ready reports automatically with Make.
- Assisted in resolving a password-change security gap where unlimited wrong-password attempts were allowed; proposed **exponential backoff on failed attempts** to throttle repeated guessing.

Customer Service Representative III
Mar 2022 - Mar 2023
Qualfon
Dumaguete City

- Handled cancellation calls in the Retention department of a **cybersecurity account**, **retaining 20-25% of cancelling customers month over month** by translating complex security concepts into clear guidance.
- Upheld **secure communication and data-handling practices** in every customer interaction, safeguarding sensitive customer data.
- Proactively brought **process improvement recommendations** to supervisors, strengthening team collaboration.

Web3 Operations Manager
Aug 2021 - Jan 2022
AMD (Axie Infinity Guild)
Marikina City

- **Promoted from scholar to guild operations manager** following a security breach, assuming full ownership of daily operations.
- **Scaled the team from 3 to 13 scholars (+333%)** by building a hiring pipeline with candidate screening, structured interviews, and live hands-on skill assessments prior to onboarding.
- **Led and mentored scholars** on strategy, communication, and digital assets, **boosting team performance by 70%**.

Projects

Northstar – AI-Assisted Kanban Task Board – Software Engineer
github.com/Frankuccino/northstar
TypeScript · React · TanStack Query · Node.js · Express.js · PostgreSQL · OpenAI SDK (Groq) · SSE · Vitest · Supertest 2026 - Present

- Built a **ReAct-style agent loop** with OpenAI SDK tool calling (Groq gpt-oss-120b) to run natural-language board operations, streaming live agent activity (tool, arguments, result, latency) to the frontend over **Server-Sent Events**.
- Enforced a **server-authoritative Kanban state machine** (WIP limits, legal transitions) so AI tool calls are treated as untrusted input and cannot create invalid board states.
- Wrote a **96-test** Vitest + Supertest suite covering the AI chat flow with a deterministic stub provider.

Video Streaming Microservices Platform – Software Engineer
github.com/Frankuccino/video-streaming-microservices
Node.js · RabbitMQ · Docker · Docker Compose · Terraform · AWS (EC2, ECR) · K3s 2026

- Decomposed a monolithic video service into **4 independently deployable microservices**, replacing point-to-point HTTP calls with a RabbitMQ fanout exchange so 2 downstream consumers process events concurrently without modifying the publisher.
- Provisioned AWS infrastructure via **Terraform** (ECR, EC2) and configured a self-managed K3s cluster for containerized workloads.

HR Management System – Lead Software Engineer
hrms-au39.onrender.com
Node.js · Express.js · MongoDB · Passport.js · MVC · Render 2023

- Stepped up as **team lead of 3**, driving the project from design through implementation, testing, and **database schema design** to deliver an **actively used 42-endpoint** system on schedule for about 5 municipal admin staff managing about 100 employee records.
- Built an **MVC architecture** backend with EJS server-side templating and Passport.js server-side authentication, **eliminating client-side-only route protection**; presented the architecture to 3 industry panelists.

Education & Certifications

DataCamp – AI Engineer for Developers Associate (Credential ID: AIEDA0015784021049) Nov 2026
KodeGo – Full Stack Web Development Bootcamp Aug 2023
St. Paul University Dumaguete – STEM Diploma Dec 2021